

L3

CLASS X - SCIENCE

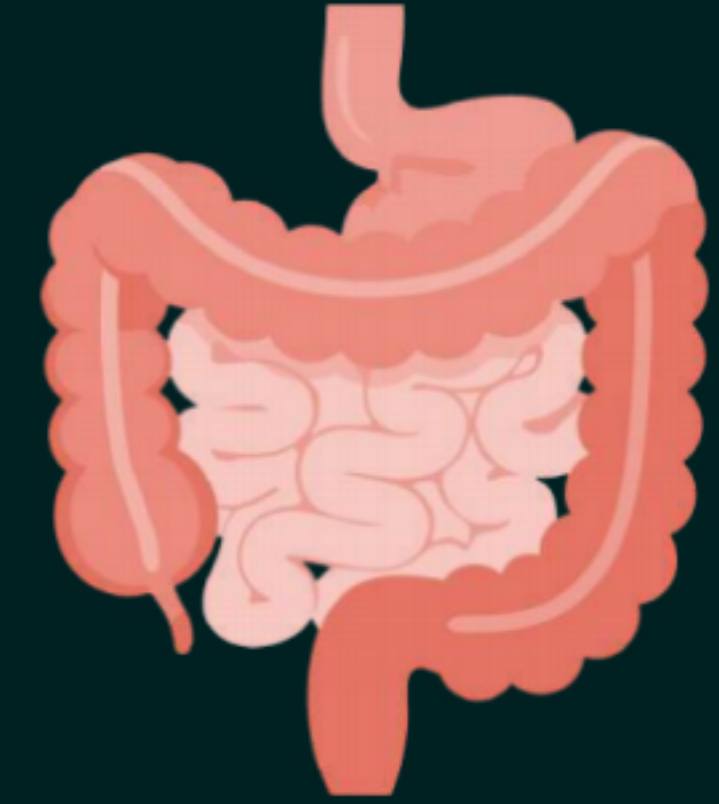
LIFE PROCESSES

PRASHANT KIRAD

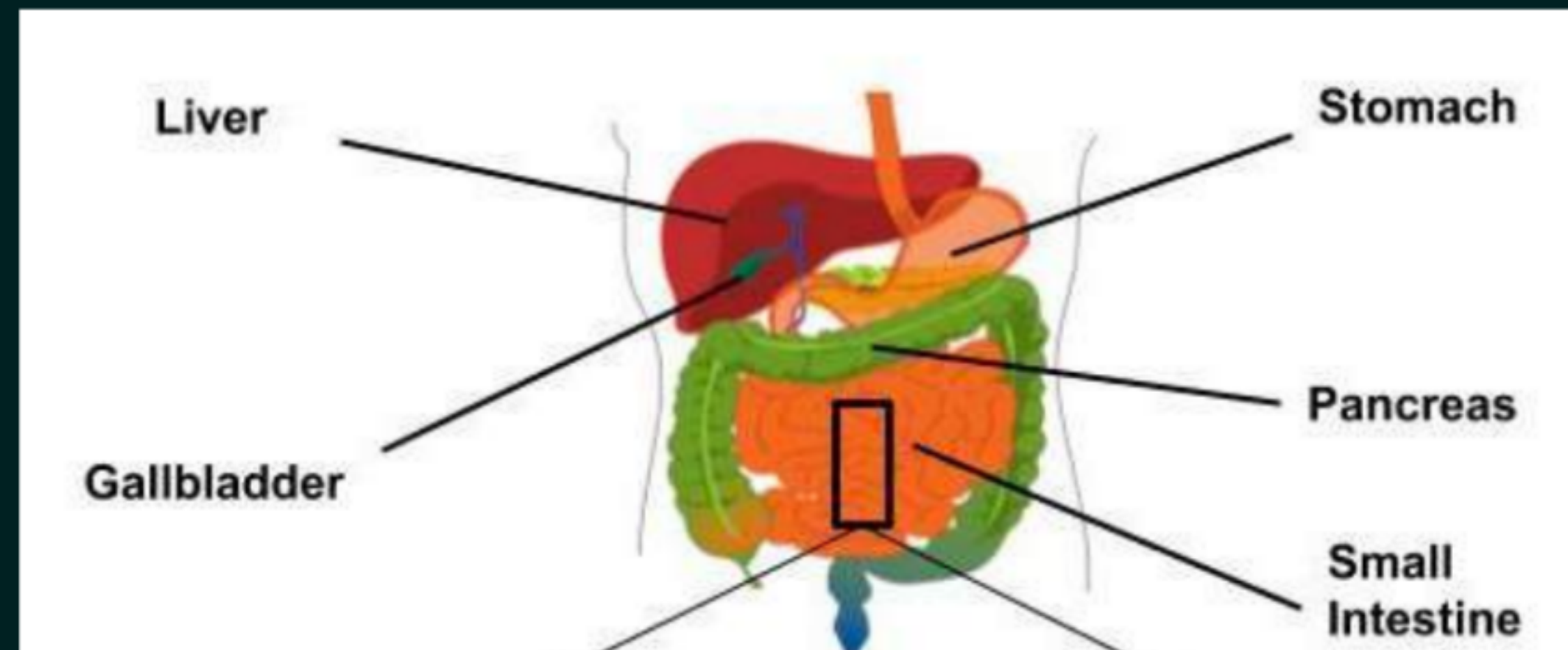
Small Intestine

(chota Don)

(5-7m)



- MCO
- The small intestine is the longest part of the alimentary canal.
 - It is the site of **complete digestion of food**.
 - The small intestine receives secretions from the liver and pancreas. Its wall also secretes **intestinal juice**, which helps in completing digestion.
 - After digestion, the food is broken down into simpler substances and is ready for absorption.



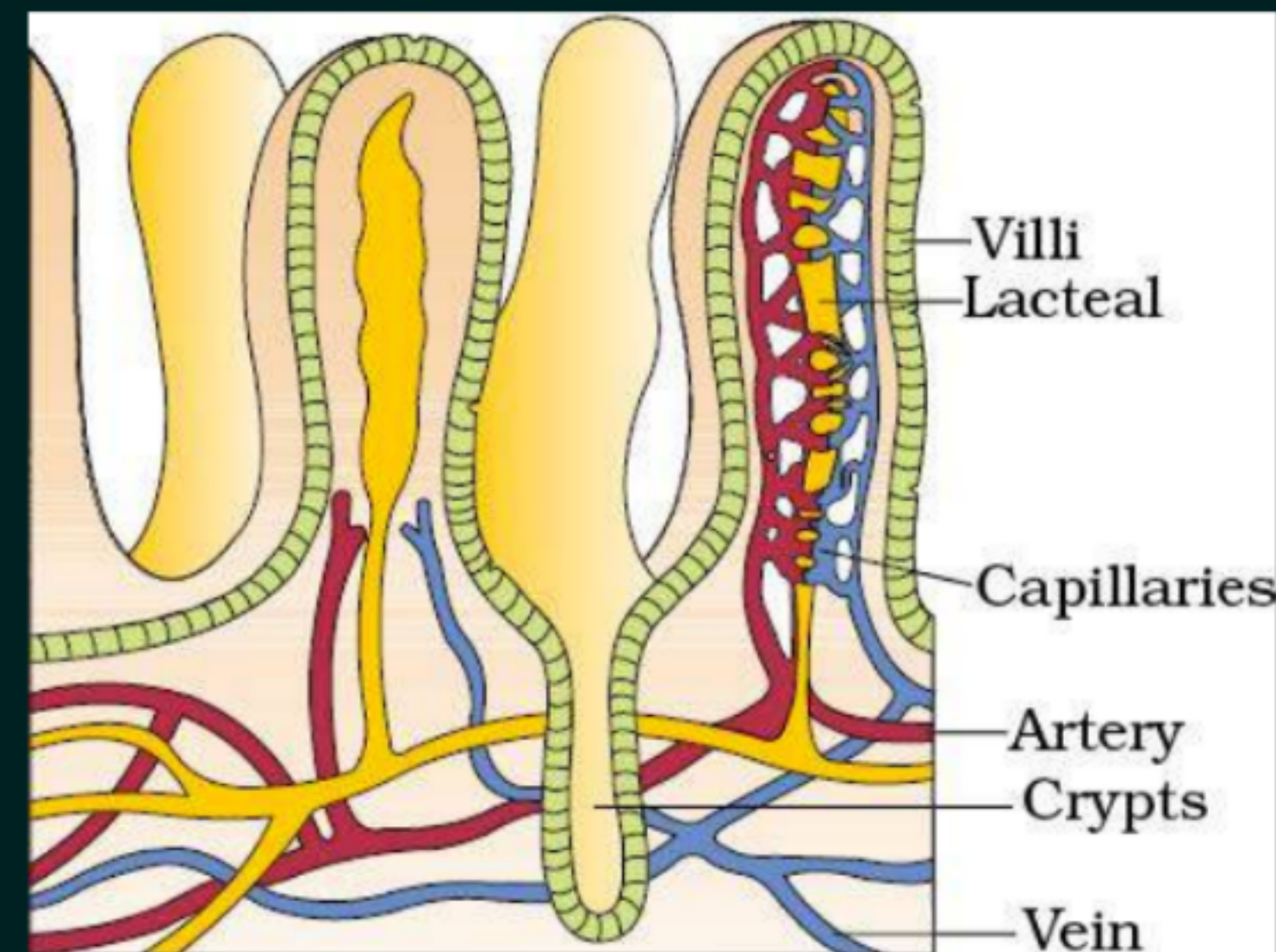
C $\rightarrow$ glucose

Protein $\rightarrow$ Amino Acids

Fats $\rightarrow$ Fatty Acid
+ Glycerol

Absorption of food in Small Intestine

- The inner lining of the small intestine has many finger-like projections called **villi**.
- Villi increase the surface area for absorption of digested food.
- Villi are richly supplied with blood vessels, which carry the absorbed food to different cells of the body.
- The absorbed food is used by the body for:

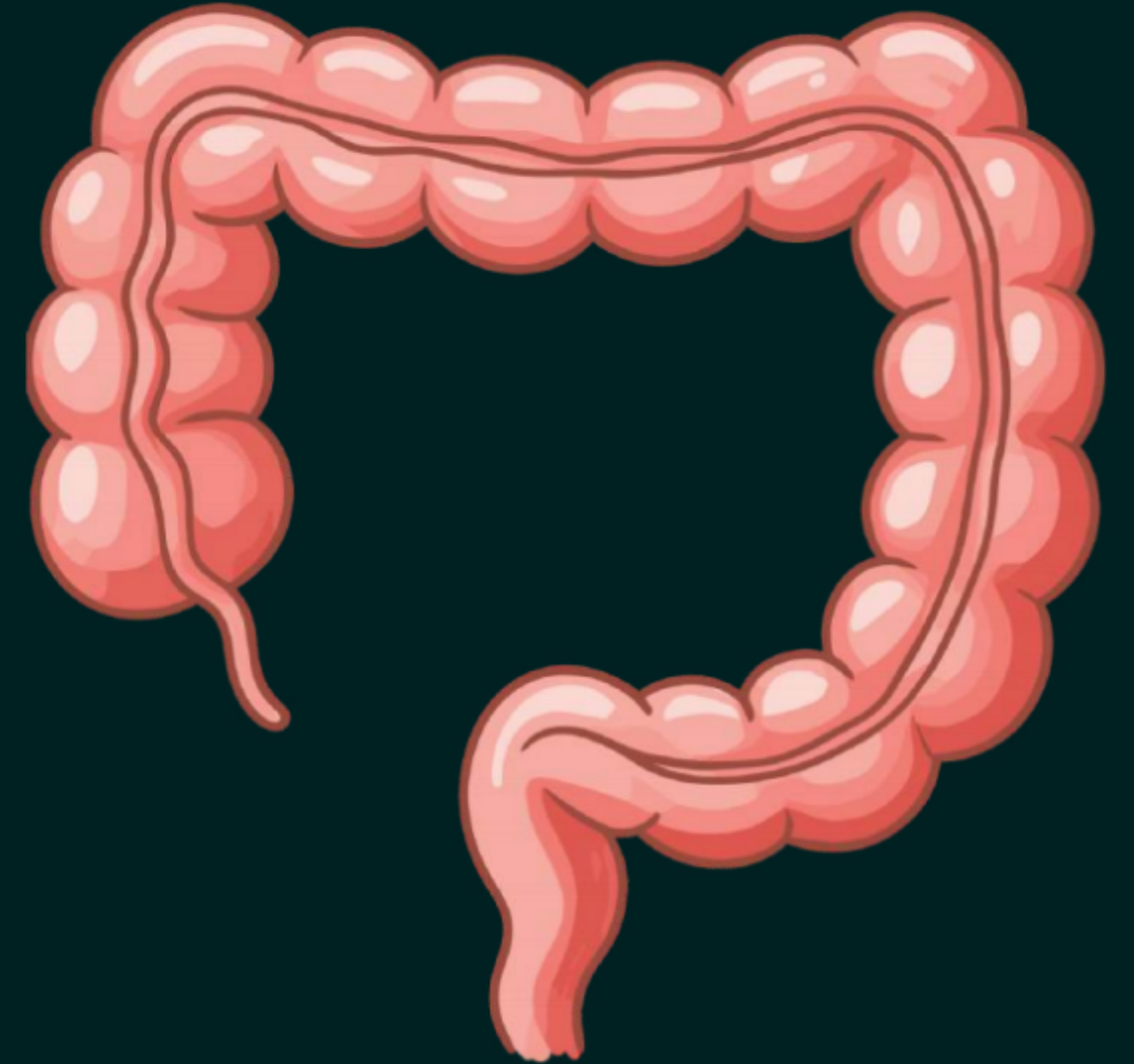


Use of Absorbed Food	Meaning
Energy	To perform life processes
Growth	To build new tissues
Repair	To repair damaged or old tissues



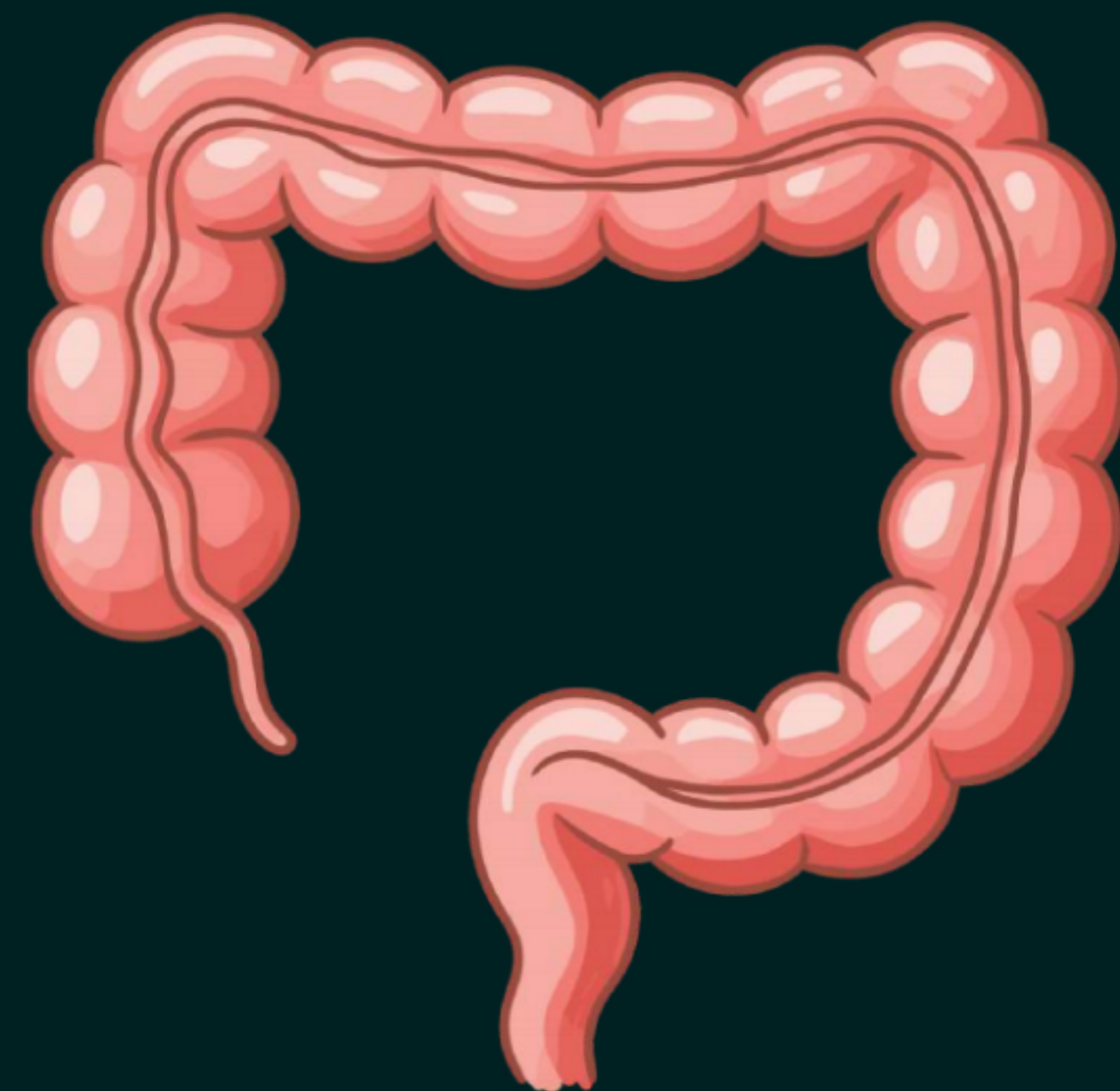
Large Intestine

- The large intestine receives the undigested and unabsorbed food from the small intestine.
- Its main function is to **absorb water from the remaining food.**
- After water absorption, the leftover waste becomes semi-solid and is passed into the rectum.
- The large intestine also contains many useful bacteria.

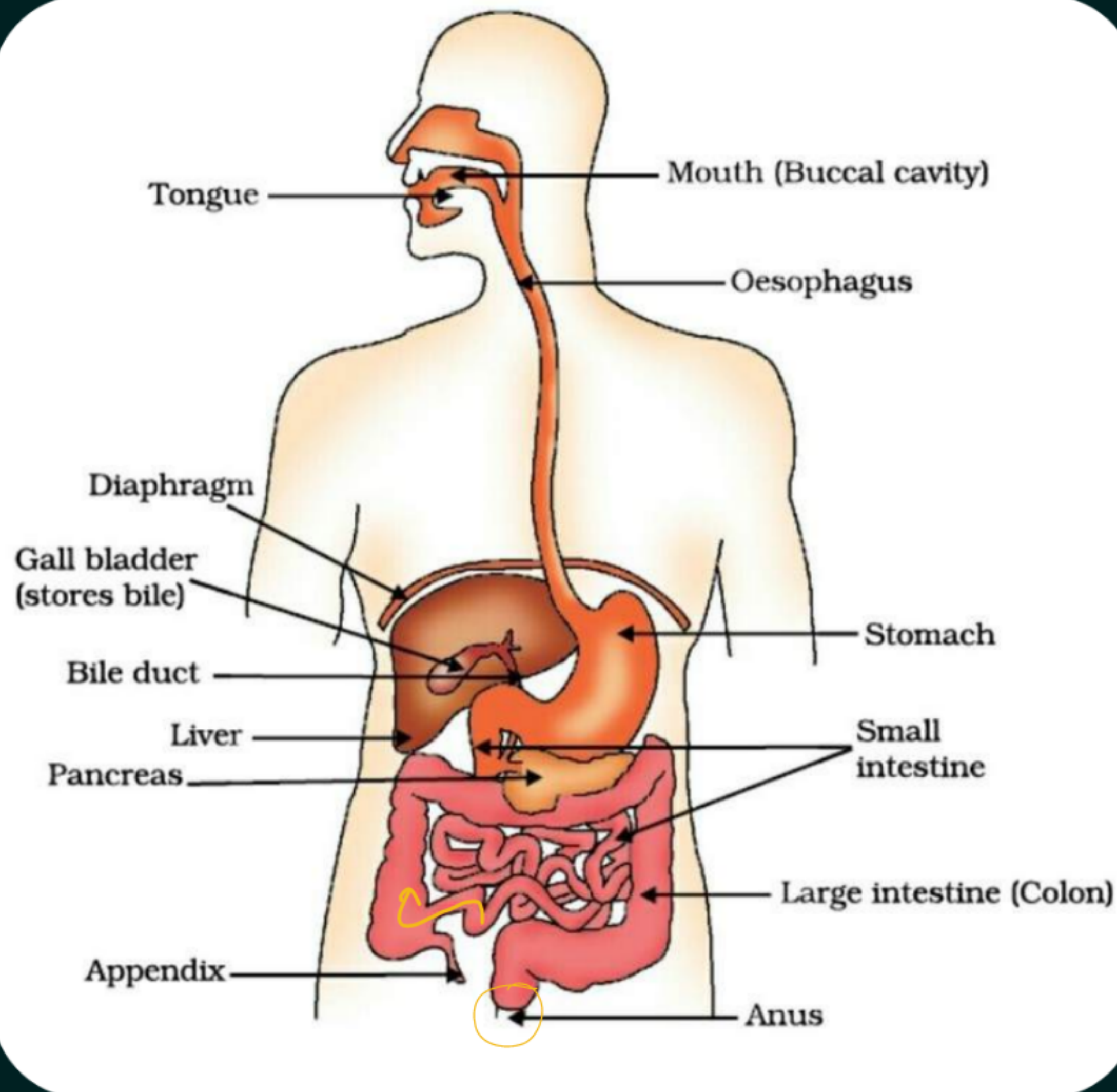


Anus

- The rectum opens outside the body through the anus.
- The anus helps in the removal of undigested food from the body.
- This process is called **egestion**.
- The anus has muscles called **anal sphincters**, which control the opening and closing of the anus during egestion.



Mouth
↓
Oesophagus
↓
Stomach
↓
Small Intestine
↓
Large Intestine
↓
Rectum
↓
Anus

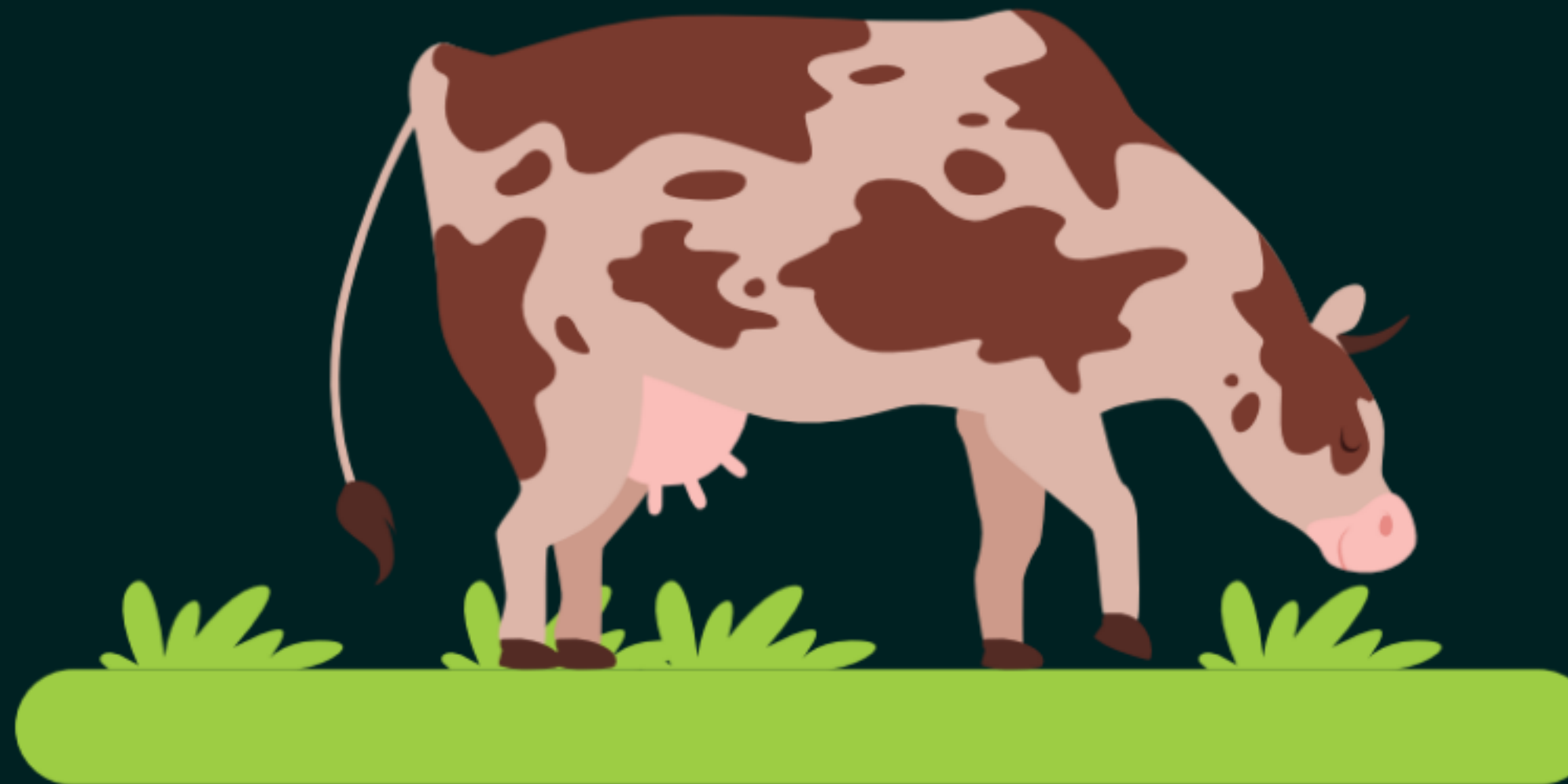


Soch ke bataao?



cellulose
digest

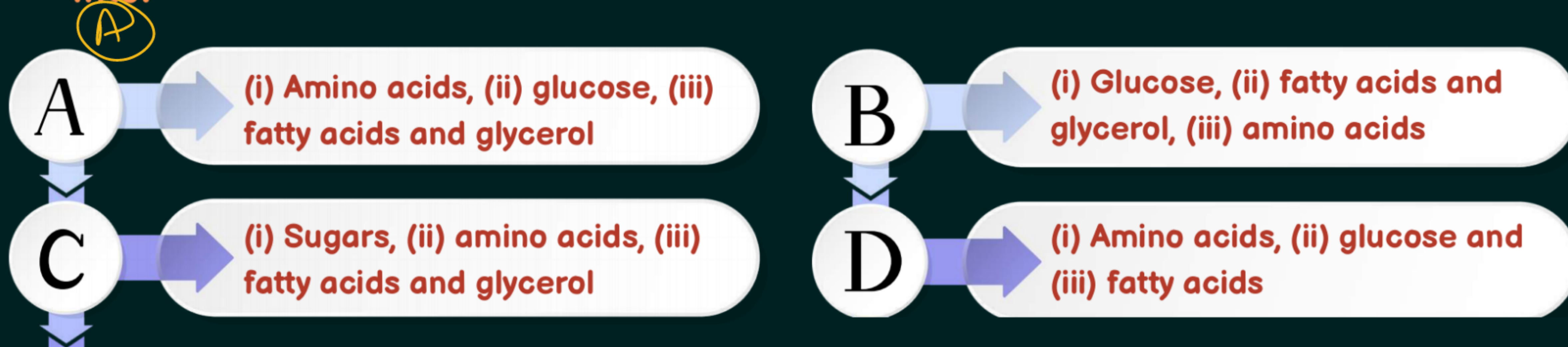
Why do herbivores have a longer small intestine than
carnivores?



LET'S TEST YOUR MEMORY



Q. In human beings, when the process of digestion is completed, the (i) proteins, (ii) carbohydrates, and (iii) fats are respectively finally converted into:



LET'S TEST YOUR MEMORY



Q. Assertion (A): The inner walls of the small intestine have finger-like projections called villi, which are rich in blood supply.

Reason (R): These villi have a large surface area to help the small intestine in completing the digestion of food.

Choose the correct option:

A

Both (A) and (R) are true and (R) is the correct explanation of (A).

B

Both (A) and (R) are true but (R) is not the correct explanation of (A).

C

(A) is true but (R) is false.

D

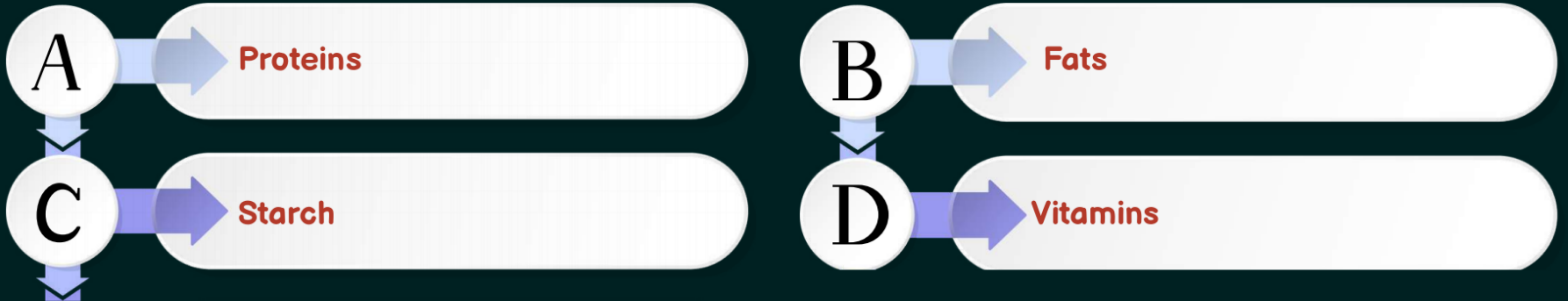
(A) is false but (R) is true.

LET'S TEST YOUR MEMORY



Q. If salivary amylase is absent, which digestion is affected first?

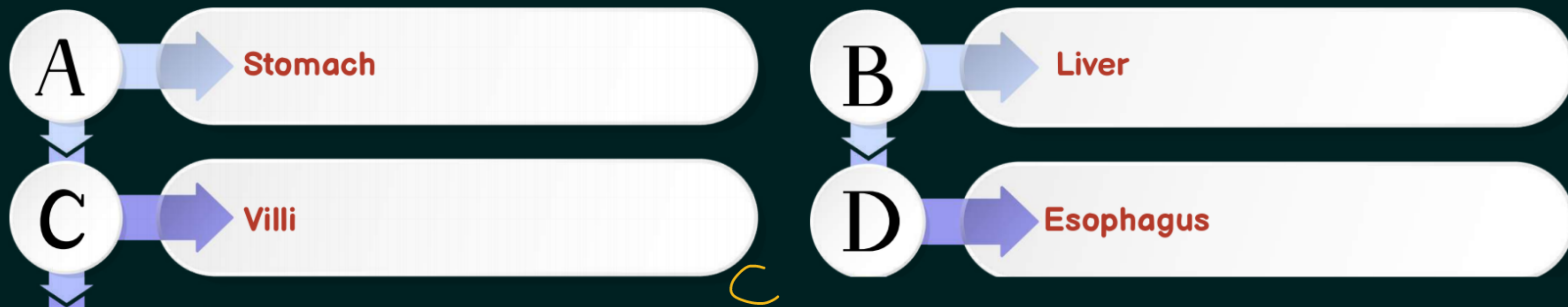
C



LET'S TEST YOUR MEMORY



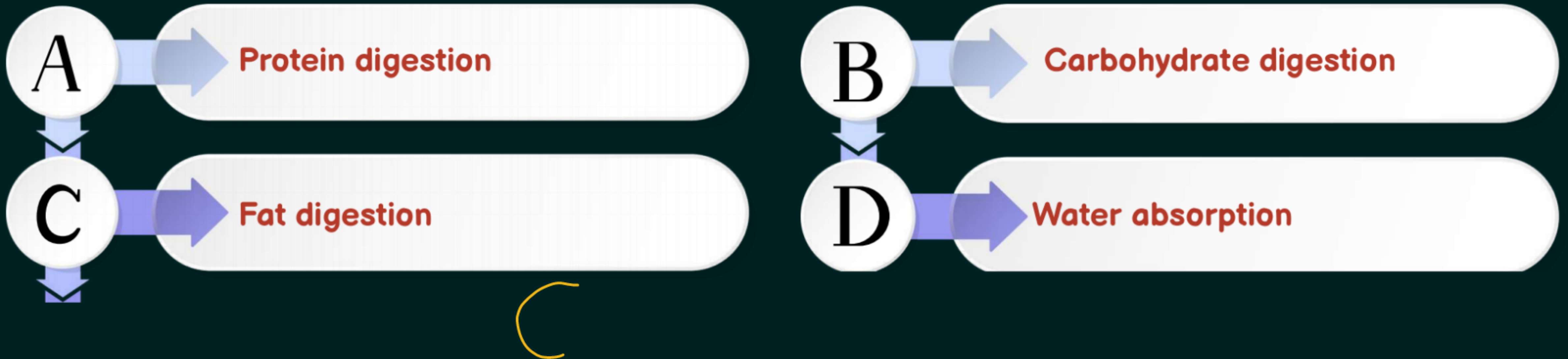
Q. A patient has normal digestion but poor absorption. Which structure is likely damaged?



LET'S TEST YOUR MEMORY



Q. If bile is not released, which process is most affected?

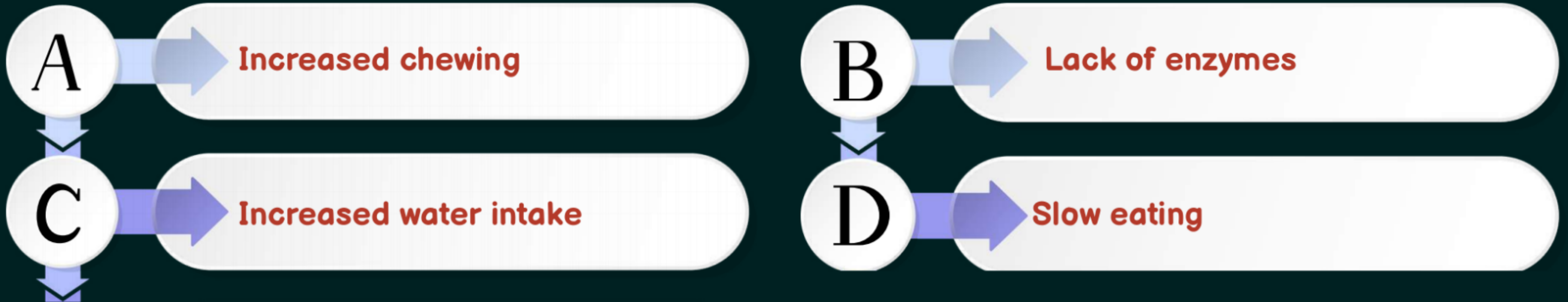


LET'S TEST YOUR MEMORY



Q. Which situation will reduce digestion efficiency the most?

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LET'S TEST YOUR MEMORY



Q. Assertion (A): The inner walls of the small intestine have finger-like projections called villi, which are rich in blood supply.

Reason (R): These villi have a large surface area to help the small intestine in completing the digestion of food.

Choose the correct option:

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (c) (A) is true but (R) is false.
- (d) (A) is false but (R) is true.

LET'S TEST YOUR MEMORY



Q. A person eats food quickly without chewing properly.

(a) Which stage of digestion is affected?

(b) How does it impact further digestion?

Answer:

(a) Mechanical digestion in mouth

(b) Larger particles reduce enzyme efficiency → slower digestion

LET'S TEST YOUR MEMORY



Q. State one role of each of the following in the human digestive system:

(i) Hydrochloric acid

(ii) Villi

(iii) Anal sphincter

(iv) Lipase

(i) Hydrochloric acid (HCl): Provides an acidic medium in the stomach to activate pepsin and kills harmful bacteria present in food.

(ii) Villi: Increase the surface area of the small intestine for efficient absorption of digested food into the blood.

(iii) Anal sphincter: Controls the opening of the anus and regulates the expulsion of faeces from the body.

(iv) Lipase: Breaks down fats (lipids) into fatty acids and glycerol.

LET'S TEST YOUR MEMORY



Q. A student takes a piece of bread and chews it slowly for 2–3 minutes. He notices that the bread starts tasting sweet.

Based on this, answer the following:

- (a) Which enzyme is present in saliva?**
- (b) What does this enzyme digest?**
- (c) Why does bread taste sweet after chewing?**

Ans.

- (a) The enzyme present in saliva is salivary amylase.**
- (b) Salivary amylase digests starch present in food.**
- (c) Bread contains starch. When we chew bread for some time, salivary amylase breaks down starch into simple sugars, so the bread starts tasting sweet.**

LET'S TEST YOUR MEMORY



Q. Why does digestion in the small intestine contribute more to nutrition than the stomach?

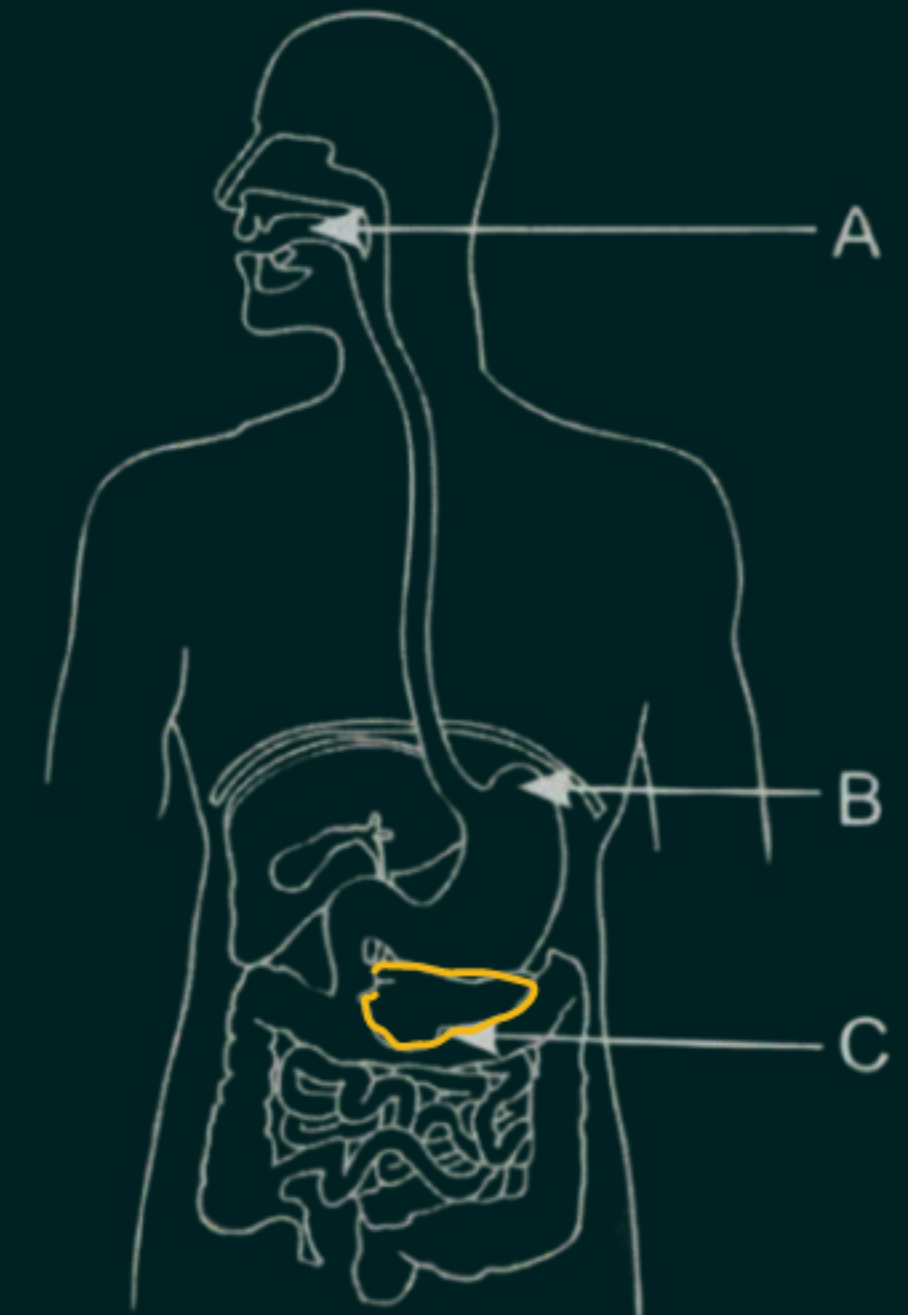
Answer:

Most digestion and absorption occur in small intestine due to enzymes and villi.

LET'S TEST YOUR MEMORY



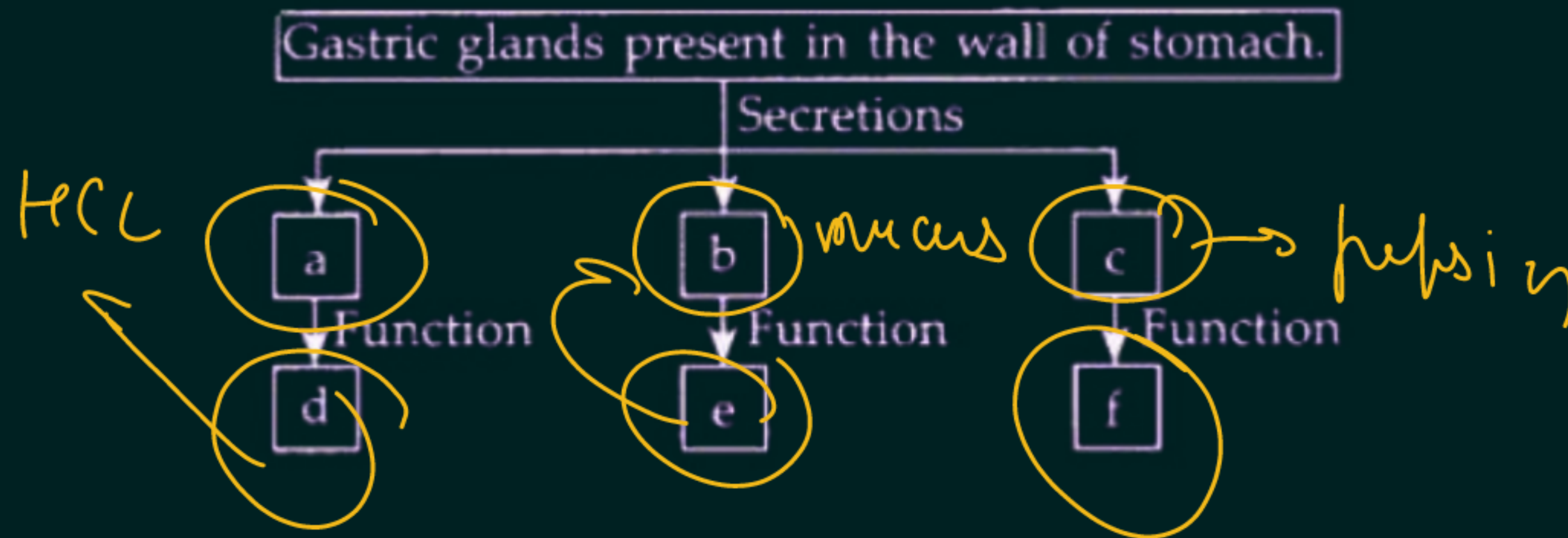
Q. (i) Study the diagram below of alimentary canal of man and name the parts marked as A, B and C. Write the name of the enzyme present in each labelled part.



LET'S TEST YOUR MEMORY



Q. Complete the following flow chart as per the given instructions.



LET'S TEST YOUR MEMORY

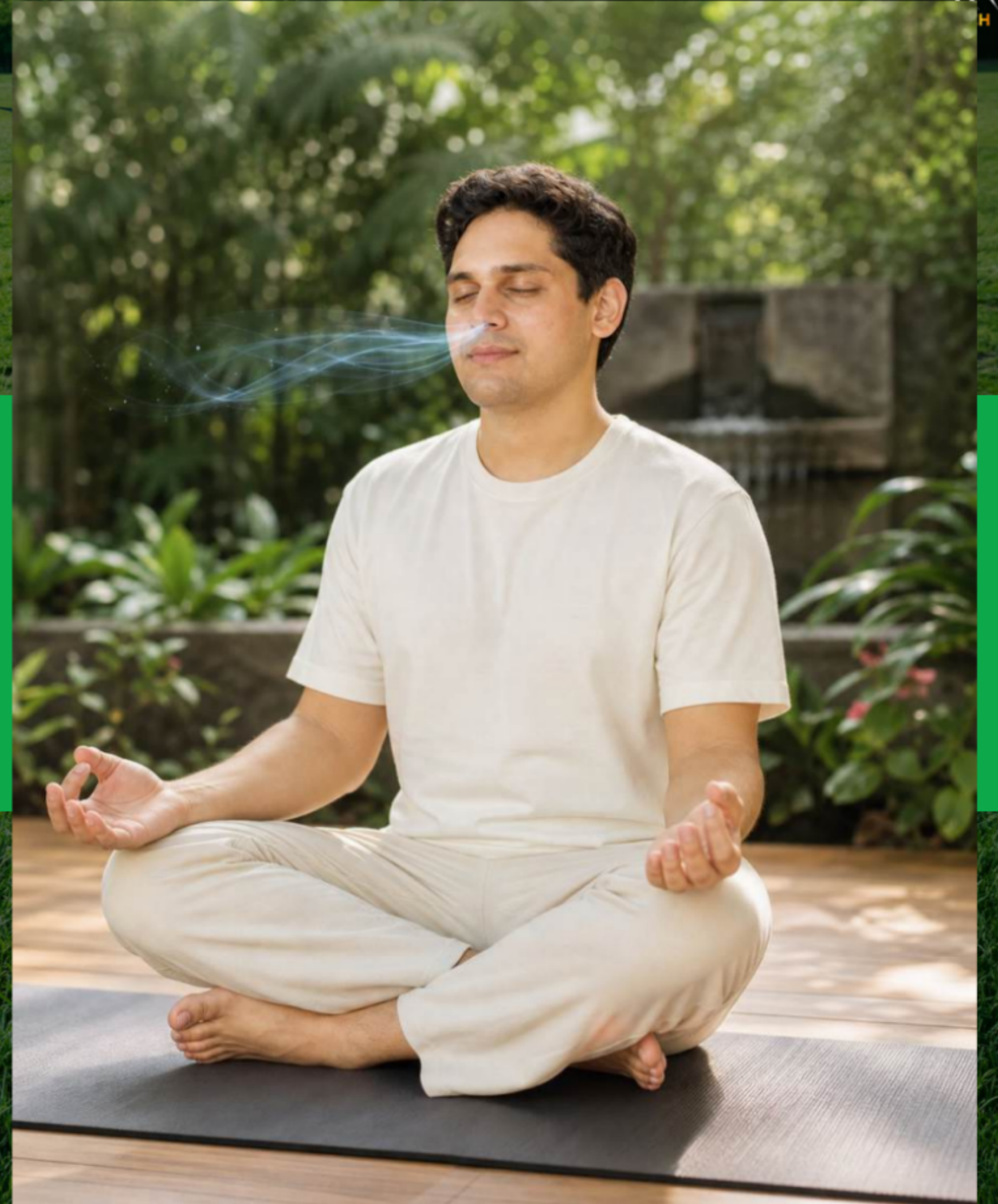


Q. A nutritionist conducted an experiment on two groups of students. Group A was given a meal rich in carbohydrates (rice, bread, sugar) and Group B was given a meal rich in proteins and fats (egg, fish, butter). After eating, the movement of food and its digestion were tracked in both groups, starting from the mouth and ending at excretion. The nutritionist made note of which enzymes and digestive organs were actively involved at each stage for each food type.

- **Group A (carbohydrate-rich):** Rice, bread, sugar
- **Group B (protein- and fat-rich):** Egg, fish, butter
- **Digestive organs:** Mouth, stomach, small intestine, large intestine
- **Digestive enzymes:** Salivary amylase, pepsin, trypsin, lipase

(A) Based on the given data, describe the pathway of carbohydrate digestion (Group A) and protein/fat digestion (Group B) through the digestive system.

Breathing and Respiration



Breathing

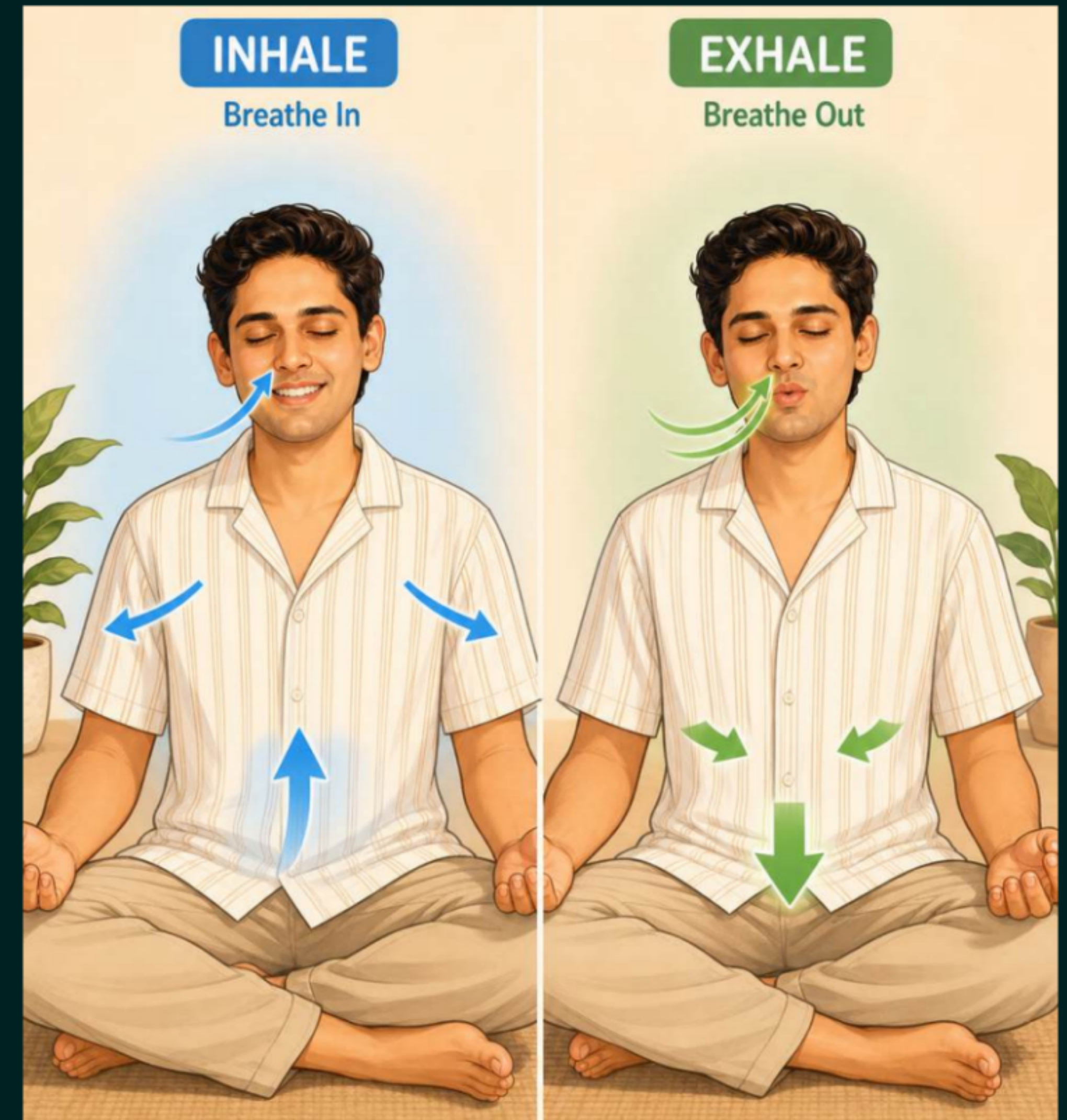
Breathing is the process of taking in oxygen-rich air and giving out carbon dioxide-rich air.

Steps of Breathing:

1. **Inhalation** – Taking air into the lungs.
2. **Exhalation** – Releasing air out of the lungs.

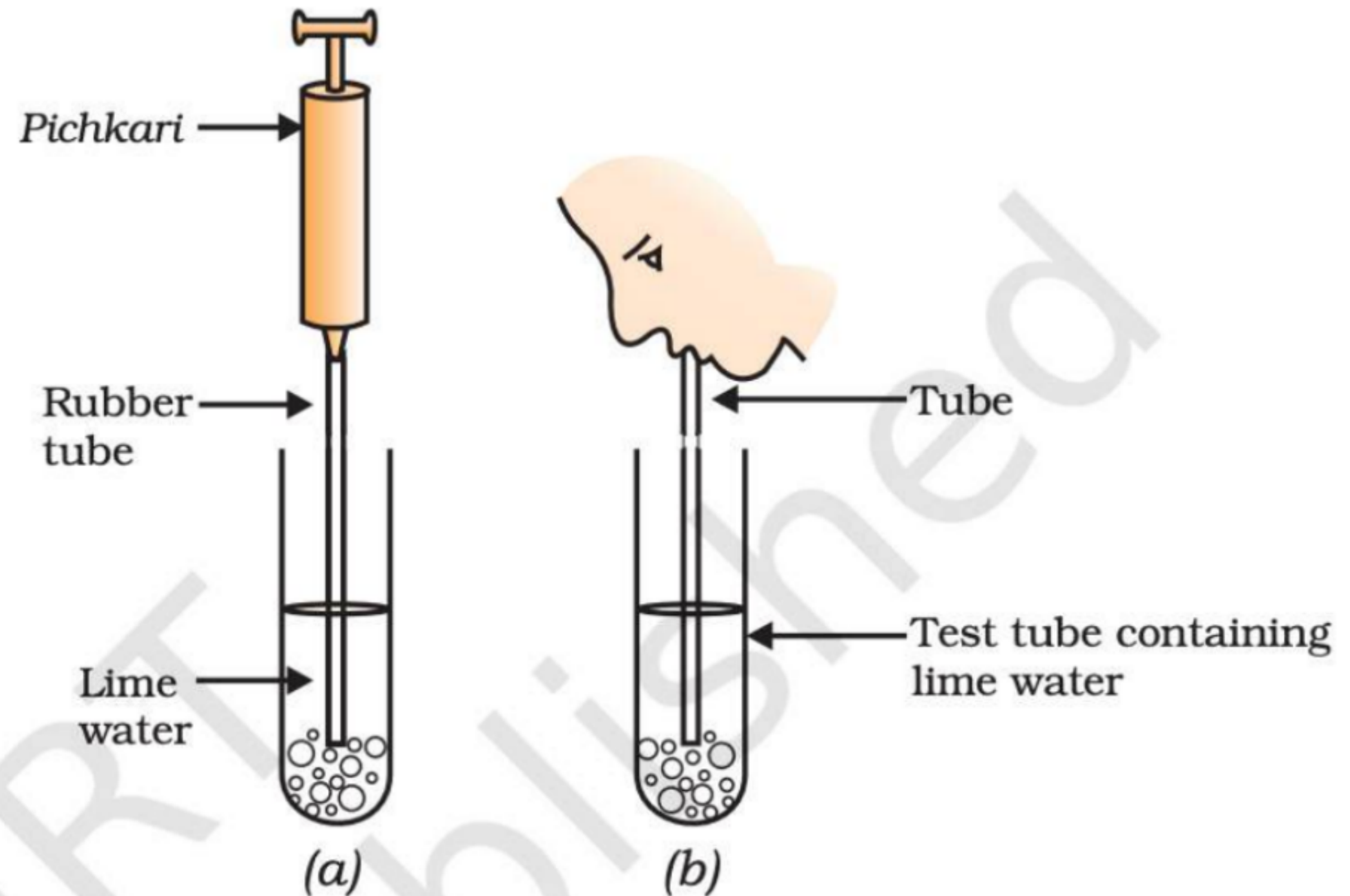
Why do we breathe?

We breathe to take in oxygen and remove carbon dioxide. Oxygen helps release energy from food, while carbon dioxide is removed as waste.



Activity 5.4

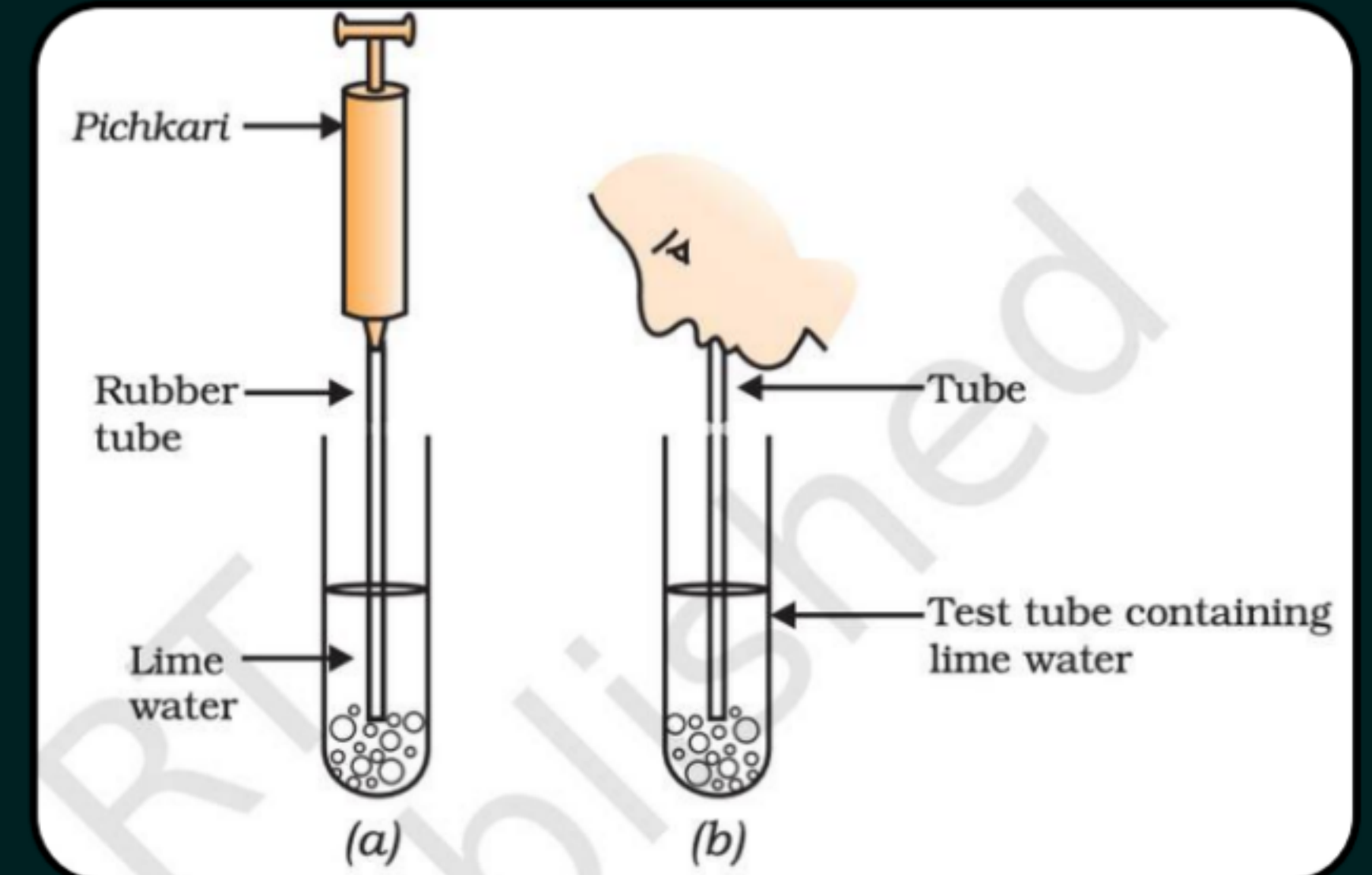
- Take some freshly prepared lime water in a test tube.
- Blow air through this lime water.
- Note how long it takes for the lime water to turn milky.
- Use a syringe or *pichkari* to pass air through some fresh lime water taken in another test tube (Fig. 5.7).
- Note how long it takes for this lime water to turn milky.
- What does this tell us about the amount of carbon dioxide in the air that we breathe out?



PK Points:

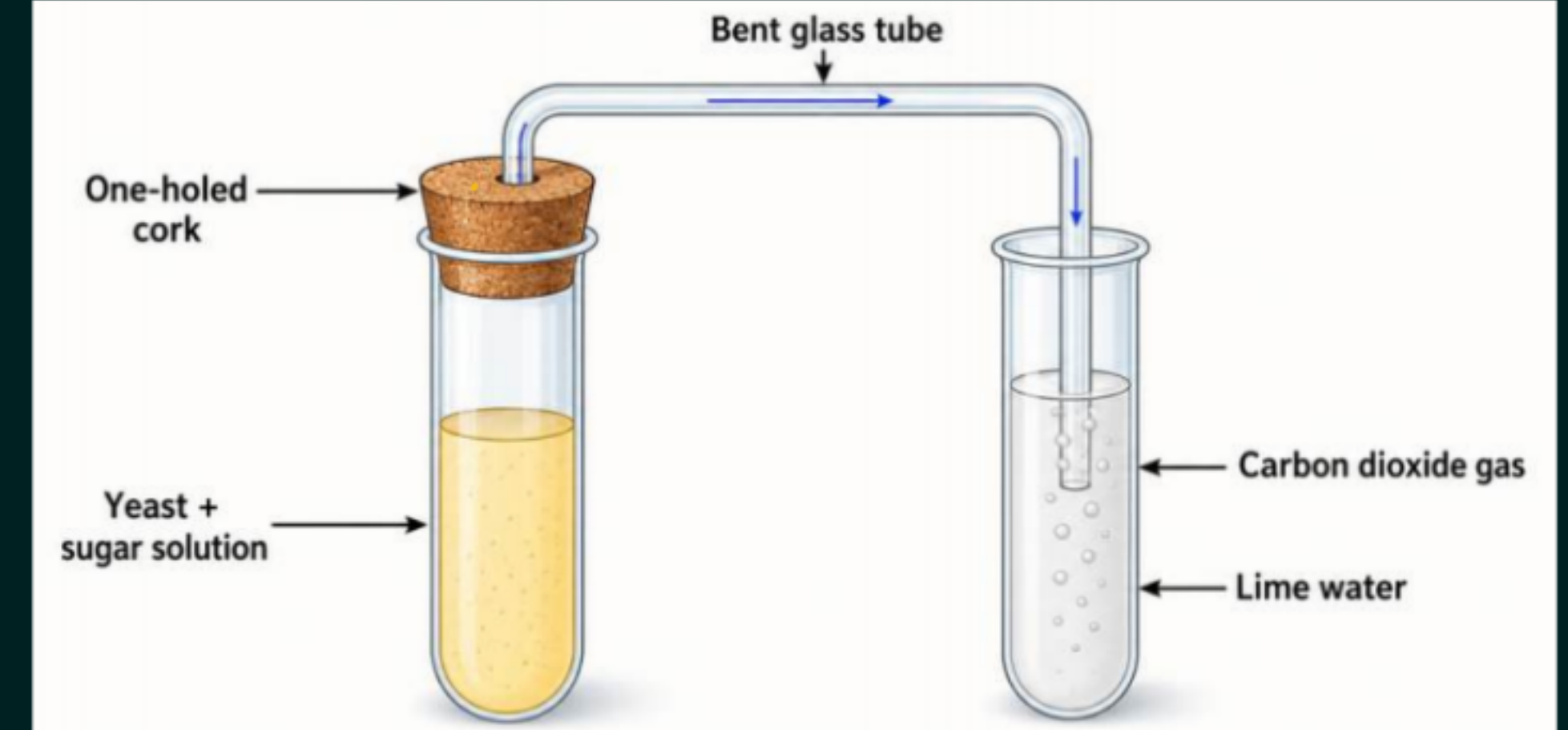


- Fresh lime water is taken in a test tube.
- Air is blown through lime water using mouth.
- Lime water turns milky quickly when we blow exhaled air into it.
- In another test tube, air is passed using a syringe/pichkari.
- This lime water also turns milky, but takes more time.
- Lime water turns milky due to the presence of carbon dioxide.
- Exhaled air contains more carbon dioxide than normal air.
- The air we breathe out is rich in carbon dioxide because food is broken down in cells during respiration.
- This activity shows that carbon dioxide is released during respiration.
- Conclusion: **Exhaled air contains more CO_2 than inhaled air.**



Activity 5.5

- Take some fruit juice or sugar solution and add some yeast to this. Take this mixture in a test tube fitted with a one-holed cork.
- Fit the cork with a bent glass tube. Dip the free end of the glass tube into a test tube containing freshly prepared lime water.
- What change is observed in the lime water and how long does it take for this change to occur?
- What does this tell us about the products of fermentation?



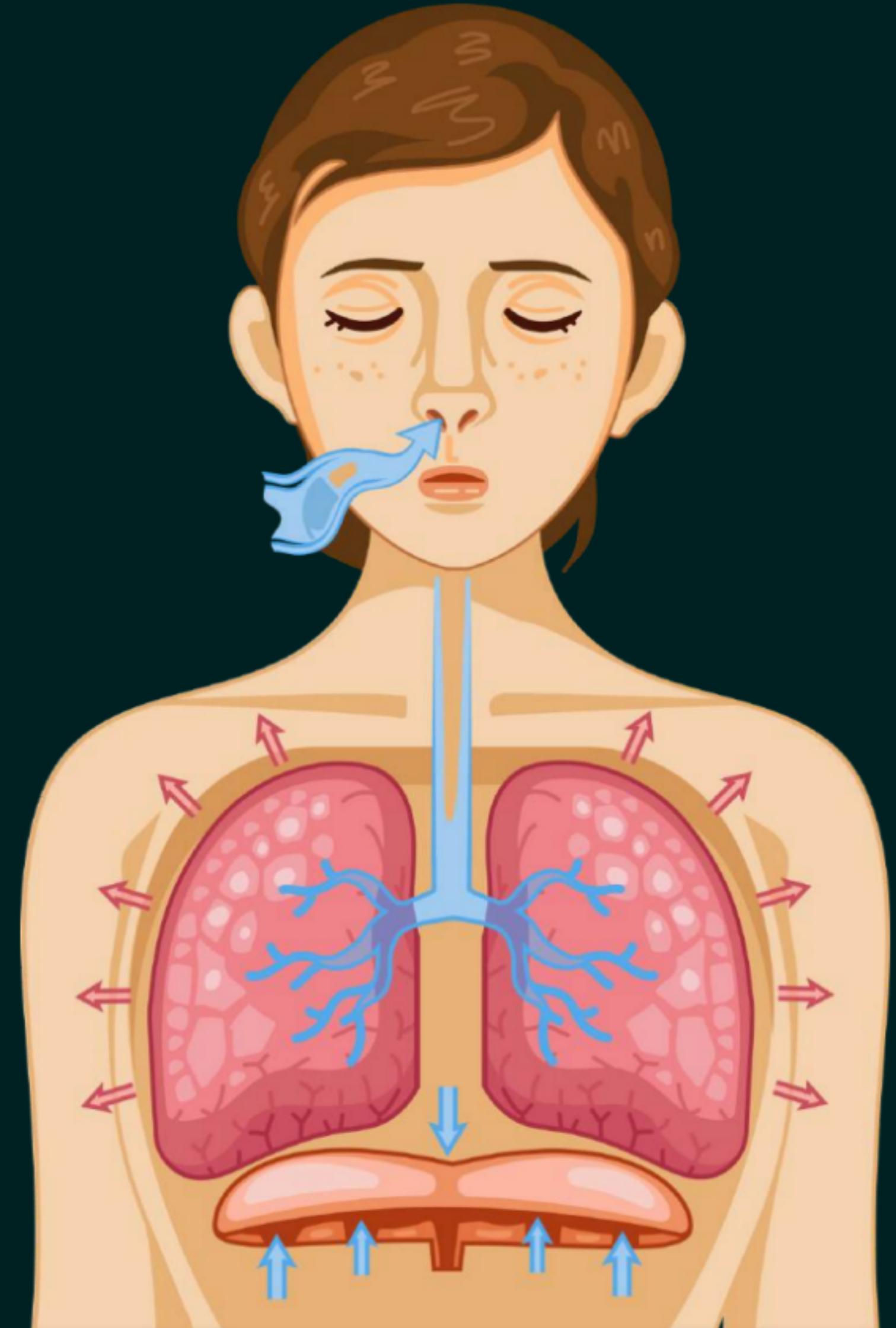
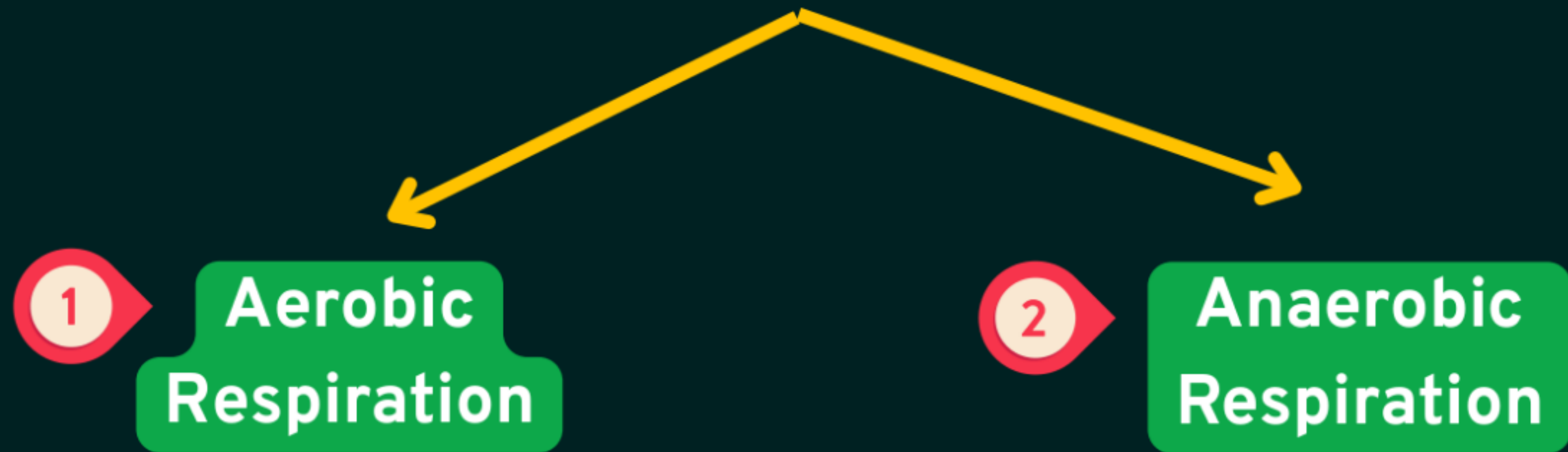
Sugar + Yeast

- Fruit juice or sugar solution is taken because yeast needs sugar as food.
- Yeast is added to the sugar solution.
- The test tube is fitted with a cork and bent glass tube to pass the gas produced into lime water.
- The free end of the glass tube is dipped in fresh lime water to test the gas produced.
- Yeast breaks down sugar in the absence of oxygen.
- This process is called **fermentation**.
- During fermentation, carbon dioxide gas is released.
- The released CO_2 passes into lime water.
- Lime water turns milky due to carbon dioxide.
- This shows that **carbon dioxide is one of the products of fermentation**.

Respiration

Respiration is the process in which food is broken down inside the cells to release energy.

In this process, the energy stored in food is released and used by the body for different life processes.





NEXT WEEK
MILTE HAI